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Course Summary Document

Digital Law and Data Protection

Chapter 2: IT Objects & Digital Actors

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Overview: The digital ecosystem operates on two essential components: manipulated matter (IT objects and data) and entities that design, exploit, or transport them (digital actors).

1. IT Objects and Data

An IT object refers to any digital entity modeled, stored, or processed by a computer system, ranging from raw binary to advanced data structures.

1.1 Personal Data

- **Definition:** Any information relating to an identified or identifiable natural person, directly or indirectly.
- **Direct Identification:** Information directly pointing to an individual without cross-referencing (e.g., Social Security Number / NIR or National ID number).
- **Indirect Identification:** Information allowing re-identification by combining contextual data points (e.g., ZIP code 16000 + Birth Date 12/04/1985 + Job Title 'Procurement Director').

Property	Pseudonymized Data	Anonymous Data
Mechanism	Replaces direct identifiers with a pseudonym (token, salted hash).	Complete and irreversible severance of link between data and individual.
Reversibility	Reversible with a secure matching table or key.	Irreversible by any reasonably deployable means.
Legal Status	Remains Personal Data (requires protection).	Loses qualification as Personal Data.

1.2 Content (Payload) vs. Metadata

- **Content (Payload):** The core substance of the transmitted or stored message intended for the end recipient (e.g., email text, attached PDF contract).
- **Metadata:** Data about the data—contextual, routing, volumetric, timing, and location information (e.g., VoIP call logs showing who called whom, call duration, timestamp, and cell tower ID, without recording audio).

1.3 Network Traces: MAC Addresses, IP Addresses, and Connection Logs

- **MAC Address (Media Access Control):** Unique 48-bit physical identifier burned into the Network Interface Card (NIC). Operates at OSI Layer 2 for local network routing.
- **IP Address (Internet Protocol):** Logical identifier for OSI Layer 3 network routing. Can be Static (fixed, permanent for servers) or Dynamic (temporary, assigned by DHCP/ISP).
- **Connection Logs:** Chronological server/network journals recording event timestamp, source IP, HTTP method, URL requested, and User-Agent string.

1.4 Business IT Objects and Data Typologies

- **Source Code vs. Executable:** Source code is human-readable logic written in programming languages (e.g., script.py); Executable (binary) is compiled machine code executed directly by the CPU (e.g., app.exe).
- **Software vs. Database:** Software is a coherent suite of programs and instructions (e.g., LibreOffice, Linux OS); Database is a structured collection of independent data accessed via DBMS (e.g., PostgreSQL, MongoDB).
- **Data Typology (Operational):** • Raw Data: Unfiltered data collected at source (e.g., 1000Hz industrial telemetry sensor stream).
  • Cleaned Data: Normalized, deduplicated, and cleansed data free of technical errors.
  • Synthetic Data: Artificially generated data (e.g., by GANs) preserving real statistical properties without representing real individuals (e.g., 100k AI-generated X-rays).

2. Digital Actors

The operation of the IT ecosystem relies on an interdependent chain of actors with specific roles, technical scopes, and legal responsibilities.

2.1 Data Controller (Responsable de Traitement - RT)

- **Role & Responsibility:** The entity (legal or natural) that decides the purpose (why) and means (how) of processing personal data (e.g., a private bank launching an online credit application service).

2.2 Data Processor (Sous-traitant)

- **Role & Responsibility:** Processes personal data strictly on behalf of and according to documented instructions from the Data Controller (e.g., a digital archiving firm storing bank documents).
  *Reclassification Risk: If the processor reuses data for its own purposes (e.g., AI model training), it becomes an independent Data Controller.

2.3 Data Protection Officer (DPO)

- **Role & Responsibility:** Expert responsible for informing, advising, and monitoring operational compliance, acting as liaison between technical teams, management, and regulatory authorities (e.g., hospital DPO validating patient record access procedures).

2.4 Software Vendor (Éditeur de Logiciel)

- **Role & Responsibility:** Designs and releases software solutions. Responsible for code quality and algorithmic security, but does not necessarily access end-user data (e.g., Adobe developing Photoshop).

2.5 Cloud Hosting Provider (Hébergeur Cloud)

- **Role & Responsibility:** Maintains physical and virtual infrastructure (servers, storage, hypervisors) to ensure operational availability, uptime, and continuity (e.g., AWS, OVHcloud).

2.6 Internet Service Provider (ISP) & Telecom Operator

- **Role & Responsibility:** Acts as a technical transport intermediary ('pure pipe') carrying IP packets across networks without altering payload content (e.g., Orange, Algérie Télécom).

2.7 Shared Responsibility Model in the Cloud

Cloud adoption redistributes operational boundaries between the Provider and the Customer (Data Controller):

- **IaaS (Infrastructure as a Service):** Provider manages physical hardware, power, network & hypervisor. Customer manages OS, security patches, runtime, app code & data (e.g., AWS EC2 VM).
- **PaaS (Platform as a Service):** Provider manages hardware, OS, runtime environment & database engine. Customer manages application code & data (e.g., Google App Engine).
- **SaaS (Software as a Service):** Provider manages entire technical stack (infrastructure, OS, app, security, updates). Customer manages user access credentials & data integrity (e.g., Microsoft 365).

2.8 AI and Data Lifecycle

- **Lifecycle Stages:** 1. Raw Data Collection -> 2. Preprocessing & Cleaning -> 3. AI Model Training -> 4. Synthetic Data Generation.
- **Third-Party AI API Integration:** Integrating company remains Data Controller. AI API Provider acts as Data Processor if it processes data solely to return results without storing/reusing payloads. Reclassification occurs if the provider uses customer data to retrain its commercial models.

3. Practical Case Studies

Case Study #1: Security Audit in a Healthcare SaaS System

- **Context:** A Healthcare SaaS app allows patients to book appointments and enter symptoms. Hosted on Cloud IaaS. An audit reveals unencrypted error logs on an S3 bucket storing patient IP addresses alongside symptom descriptions.
- **Objects Analysis:** Payload = symptom descriptions (sensitive health data); Metadata = Source IP & timestamp; Combination = Highly identifying and sensitive personal data.
- **Actors Qualification:** SaaS Vendor (Data Controller) failed by misconfiguring application error logging; Cloud Provider (Data Processor) provided raw IaaS storage infrastructure and is not responsible for misconfigured customer logging.

Case Study #2: Banking Email Processing via External AI API

- **Context:** A bank automatically analyzes customer complaint emails using an external Natural Language Processing (NLP) REST API.
- **Workflow & Qualification:**
  1. Raw Email Ingestion (Bank - Controller)
  2. Local Anonymization/Preprocessing (Bank removes names & account numbers)
  3. API Call with Cleaned Payload
  4. External AI API (Processor) analyzes text and returns sentiment score without retaining customer text.